Planning with Monte Carlo tree search

Imagine many endings, then choose the first step

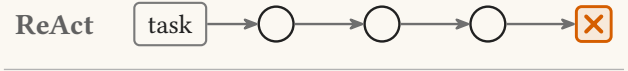
Carlos Cotrini

ETH Zurich

October 2026

Language Agent Tree Search (LATS)

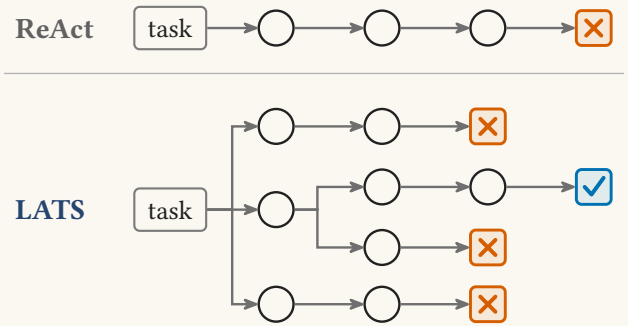
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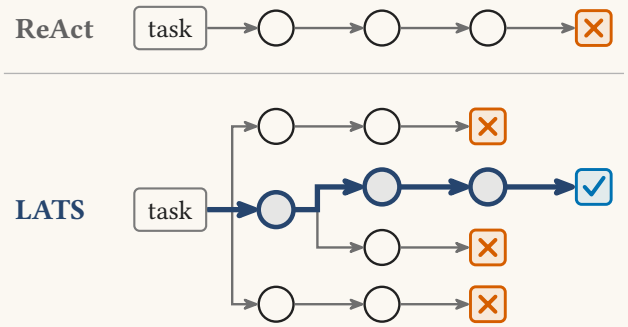
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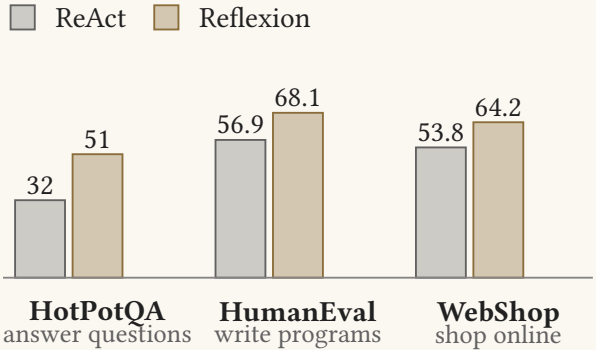
1 A plain agent, ReAct, follows one line of steps, with no way back.

2 LATS tries several steps at each fork.

3 It searches that tree by **Monte Carlo tree search**.

What the tree buys

The same model, GPT-3.5, scores higher when it searches.

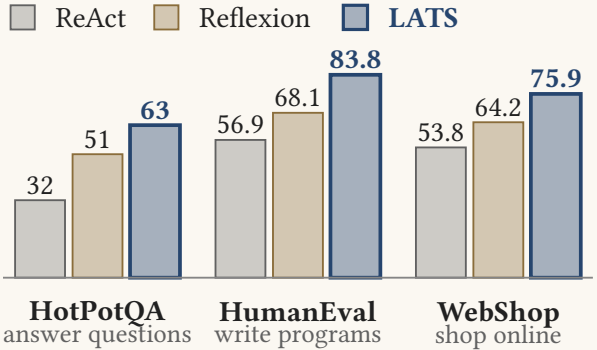


1 ReAct follows one line of steps; Reflexion retries, with notes on its failures.

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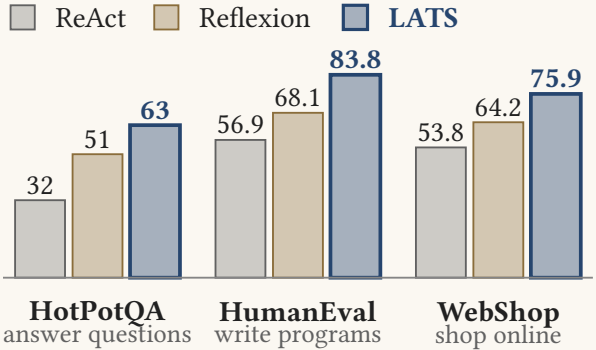
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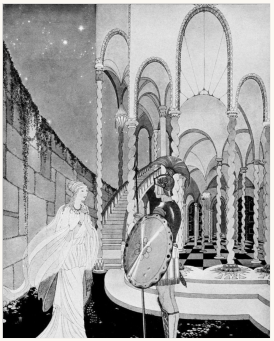
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2 LATS searches a tree of lines and leads on all three.

3 The price: more computation per task.

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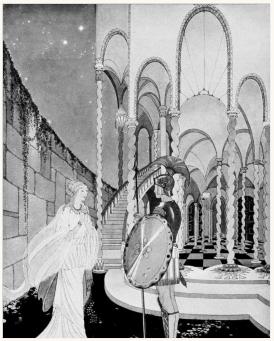
The Minotaur, and the two sails



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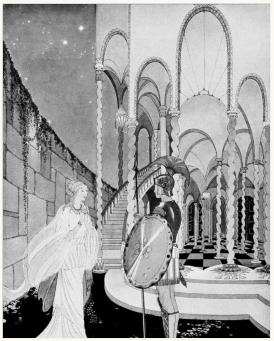


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The ship had a black sail. His father said: if you come back alive, spread white sails.

Apollodorus, *Library* 3.1.4, 3.15.8, *Epitome* 1.7 (Frazer, 1921).

The thread, and the black sail



Theseus and Ariadne, Pompeii (drawing by G. Discanno).

Ariadne asked the right question, how to get out again, and gave Theseus a ball of thread.

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He tied one end to the door, killed the monster and followed the thread out.

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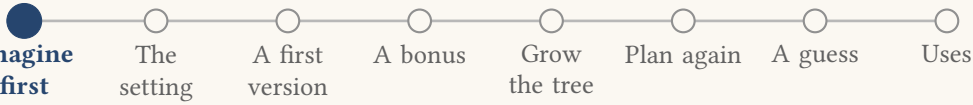
Then he sailed home and forgot the white sails.

His father Aegeus saw the black sail, supposed him dead, and threw himself down.

Apollodorus, *Epitome* 1.8 to 1.10 (Frazer, 1921).

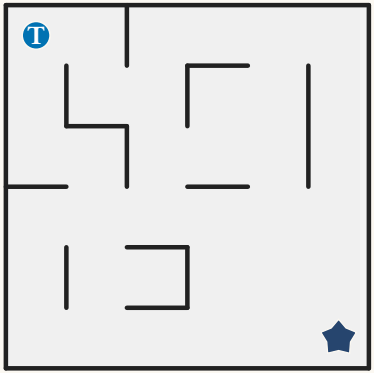
Imagine before you step

A labyrinth with two Medusas, and a habit that programs share.



Theseus and the two Medusas

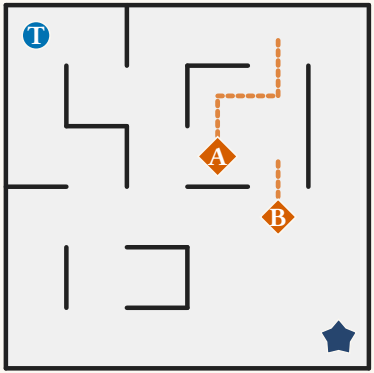
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1 Theseus, **T**, wants the lair, the star.

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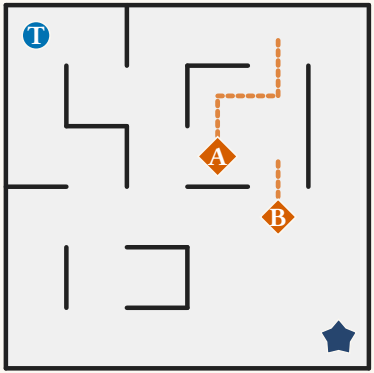


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2 Medusas **A** and **B** walk dotted rounds for ever.

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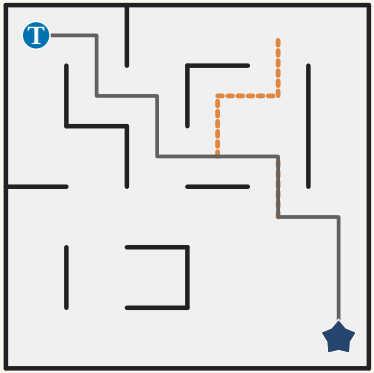
1 Theseus, **T**, wants the lair, the star.

2 Medusas **A** and **B** walk dotted rounds for ever.

3 Anyone in her cell, or the cell she enters next, turns to stone.

He can imagine the ending

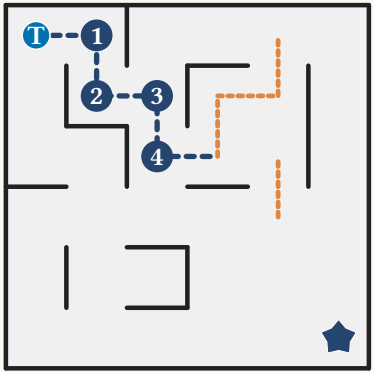
A wrong step is final, but an imagined walk costs nothing.



1 On paper the shortest route is 10 steps.

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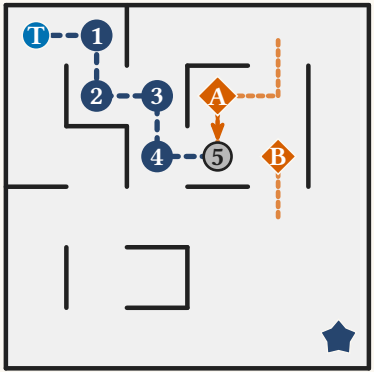


1 On paper the shortest route is 10 steps.

2 In his head he walks it, turn by turn.

He can imagine the ending

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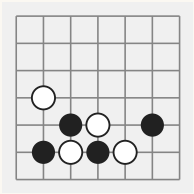
1 On paper the shortest route is 10 steps.

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3 At turn 5 Medusa **A** reaches him: stone. Nothing is lost.

The same habit runs inside programs you know

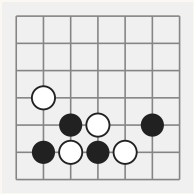
Some Go programs and web agents imagine many endings first.



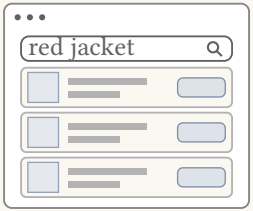
a program that plays Go

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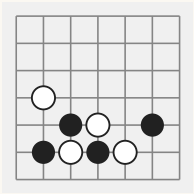
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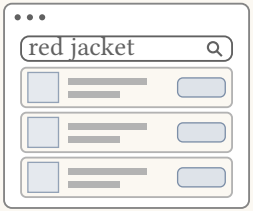
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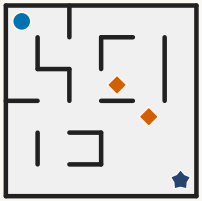
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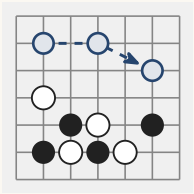
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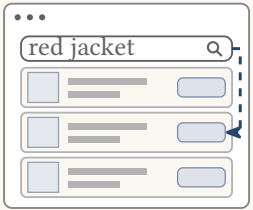
Theseus

The same habit runs inside programs you know

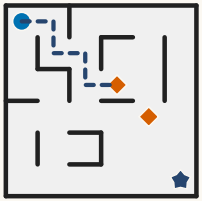
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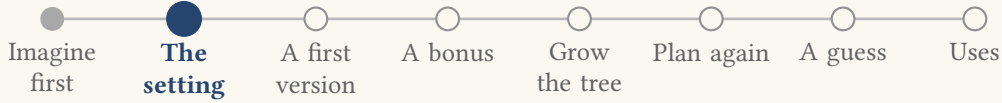


Theseus

Dashed: the endings each one imagines first.

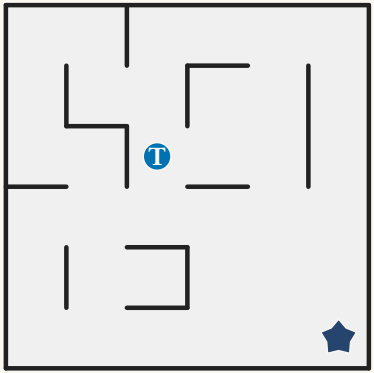
The setting

States, actions, returns and values, told on the labyrinth.



State and actions

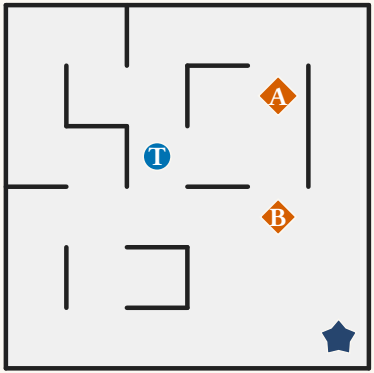
State: a cell and a turn. Action: stay, or step.



1 Here: the **T**'s cell, turn 4.

State and actions

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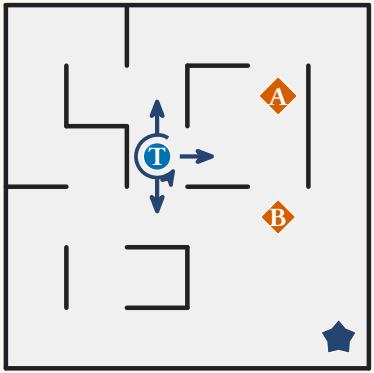


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2 The Medusas keep a timetable: another turn, another state.

State and actions

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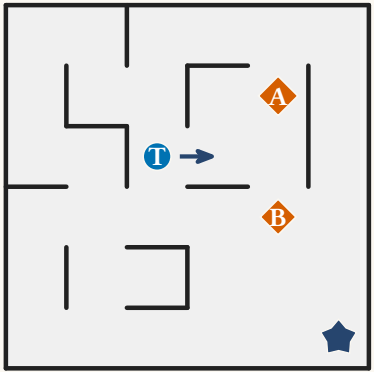
1 Here: the **T**'s cell, turn 4.

2 The Medusas keep a timetable: another turn, another state.

3 Here: four actions, stay or step up, right or down.

Transitions

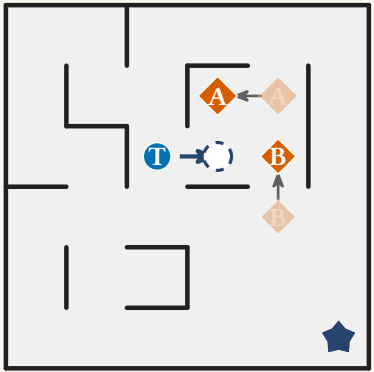
Asking the world what follows an action costs nothing.



1 He asks about one action: step right.

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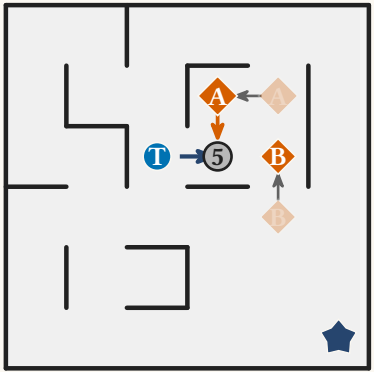


1 He asks about one action: step right.

2 The answer: the next state, turn 5.

Transitions

Asking the world what follows an action costs nothing.



1 He asks about one action: step right.

2 The answer: the next state, turn 5.

3 It also says if the run is over: stone.

Reward and return

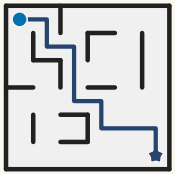
A run pays a return G, from 0 to 1: the sooner at the lair, the more.

Reward: a turn pays 0, except reaching the lair on turn t: 0.97^t.
Return G**:** the rewards added up.

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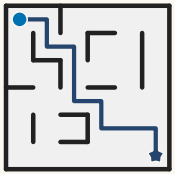


at the lair
on turn 10
$G = 0.97^{10} \approx 0.74$

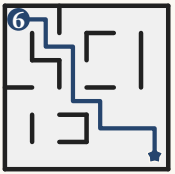
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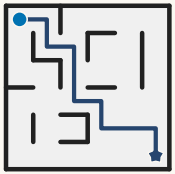


waits 6 turns, then at
the lair on turn 16
$G = 0.97^{16} \approx 0.61$

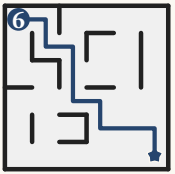
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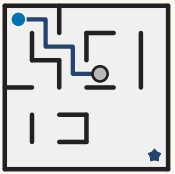
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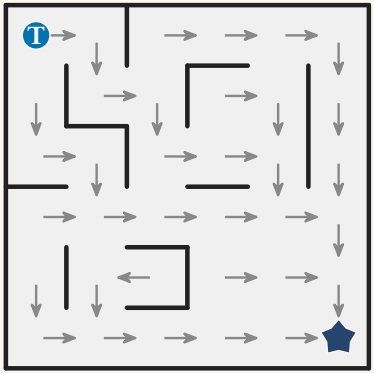
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turned to stone
on turn 5
$G = 0$

Policy and trajectory

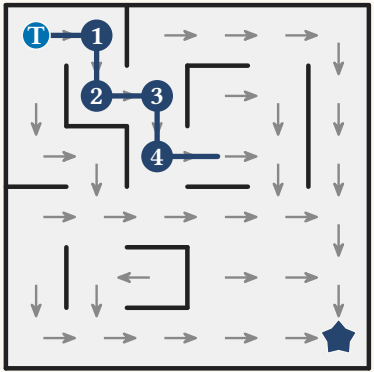
A policy picks an action in every state.



1 This one steps to the cell nearest the lair.

Policy and trajectory

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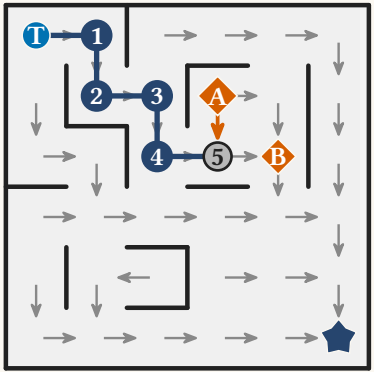


1 This one steps to the cell nearest the lair.

2 From the door it gives a trajectory.

Policy and trajectory

A policy picks an action in every state.



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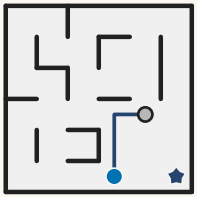
2 From the door it gives a trajectory.

3 It ends in stone on turn 5: $G = 0$.

The value of a state

A state's value $V(s)$ is the average G over many runs.

Five random runs from the blue dot, on turn 6.



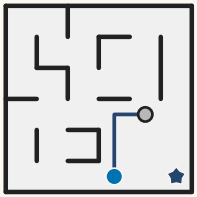
stone on turn 9

$G = 0.00$

The value of a state

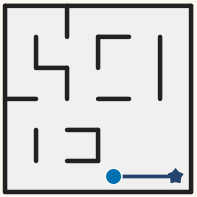
A state's value $V(s)$ is the average G over many runs.

Five random runs from the blue dot, on turn 6.



stone on turn 9

$G = 0.00$



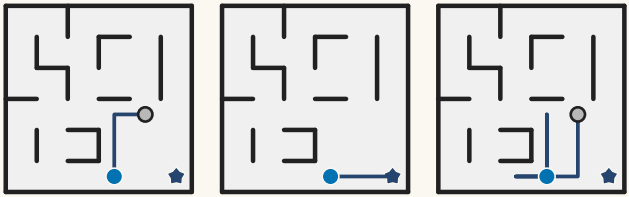
lair on turn 10

$G = 0.74$

The value of a state

A state's value $V(s)$ is the average G over many runs.

Five random runs from the blue dot, on turn 6.



stone on turn 9
$G = 0.00$

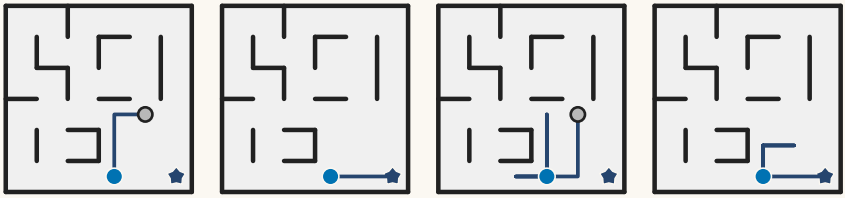
lair on turn 10
$G = 0.74$

stone on turn 15
$G = 0.00$

The value of a state

A state's value $V(s)$ is the average G over many runs.

Five random runs from the blue dot, on turn 6.



stone on turn 9
$G = 0.00$

lair on turn 10
$G = 0.74$

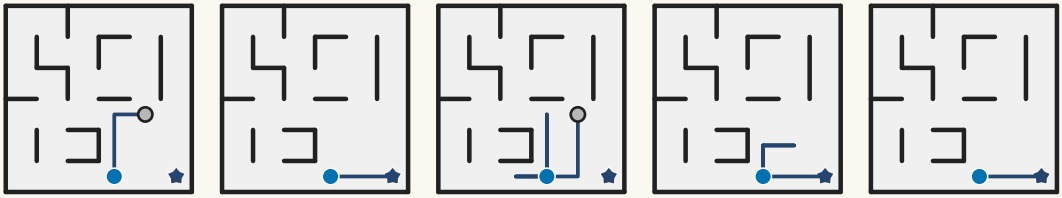
stone on turn 15
$G = 0.00$

lair on turn 18
$G = 0.58$

The value of a state

A state's value $V(s)$ is the average G over many runs.

Five random runs from the blue dot, on turn 6.

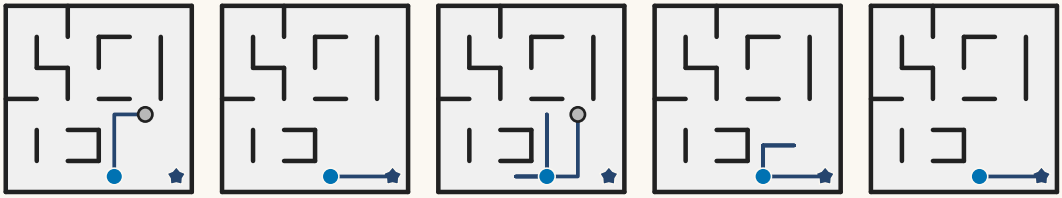


stone on turn 9	lair on turn 10	stone on turn 15	lair on turn 18	lair on turn 8
$G = 0.00$	$G = 0.74$	$G = 0.00$	$G = 0.58$	$G = 0.78$

The value of a state

A state's value $V(s)$ is the average G over many runs.

Five random runs from the blue dot, on turn 6.

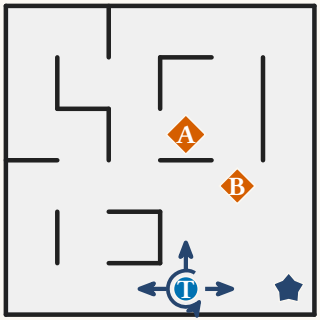


stone on turn 9	lair on turn 10	stone on turn 15	lair on turn 18	lair on turn 8
$G = 0.00$	$G = 0.74$	$G = 0.00$	$G = 0.58$	$G = 0.78$

The five average 0.42, 10,000 runs 0.24: the $V(s)$ of the policy-gradient lectures.

The goal

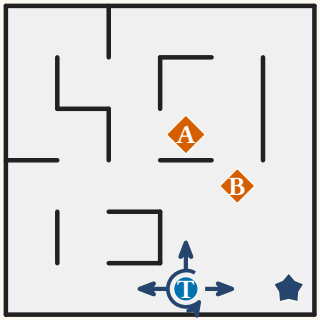
Choose the action with the largest average G.



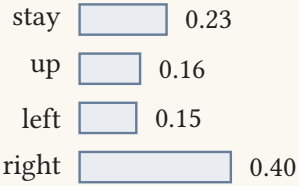
1 Per action: imagine many runs, average their G.

The goal

Choose the action with the largest average G.



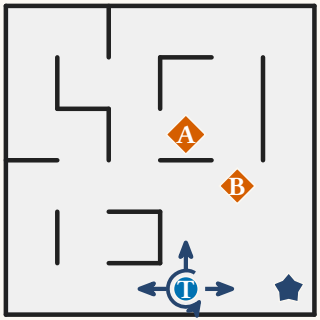
average G of 10,000 random runs after each action



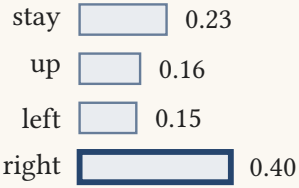
1 Per action: imagine many runs, average their G.

The goal

Choose the action with the largest average G.



average G of 10,000 random runs after each action

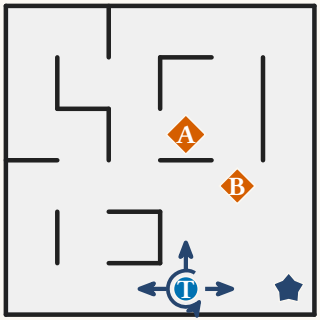


1 Per action: imagine many runs, average their G.

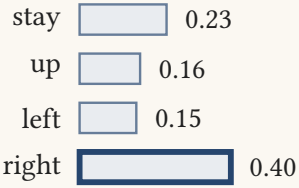
2 Play the largest: here, right.

The goal

Choose the action with the largest average G.



average G of 10,000 random runs after each action



1 Per action: imagine many runs, average their G.

2 Play the largest: here, right.

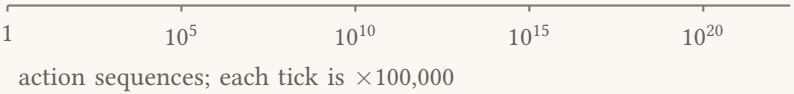
3 No training, only simulation.

Too many futures

The futures from the door explode with depth.

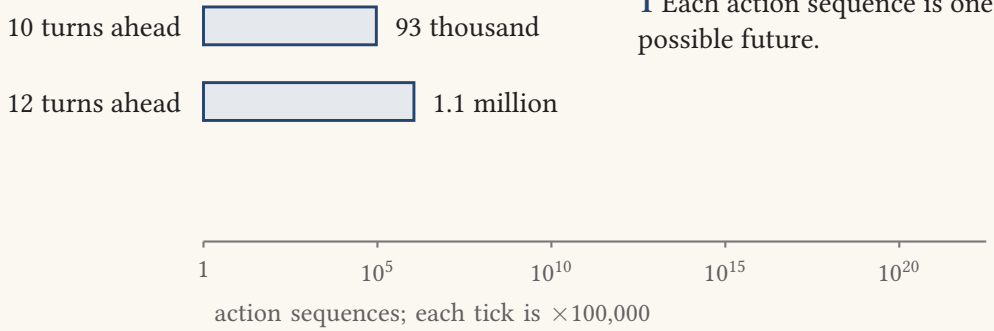
10 turns ahead  93 thousand

1 Each action sequence is one possible future.



Too many futures

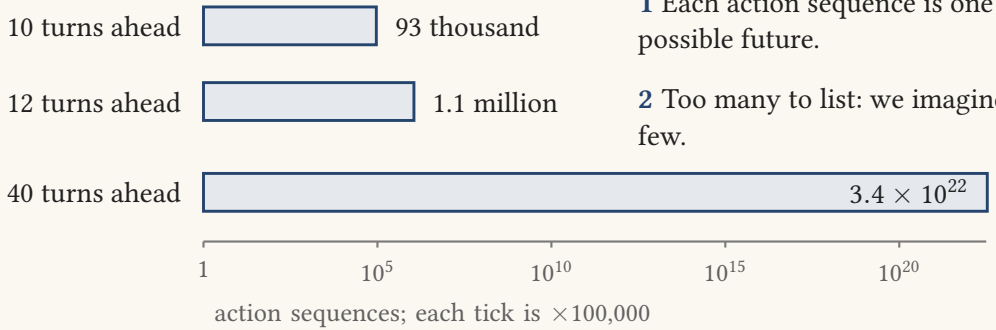
The futures from the door explode with depth.



1 Each action sequence is one possible future.

Too many futures

The futures from the door explode with depth.

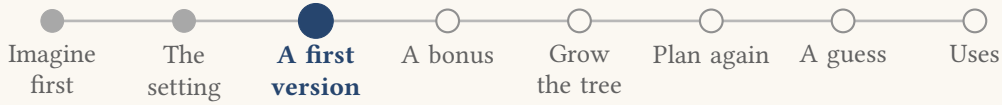


1 Each action sequence is one possible future.

2 Too many to list: we imagine a few.

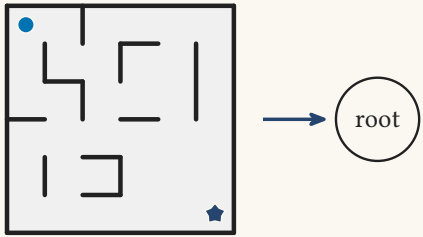
A first version

A tree of passes, and the move it chooses.



A counter on every state

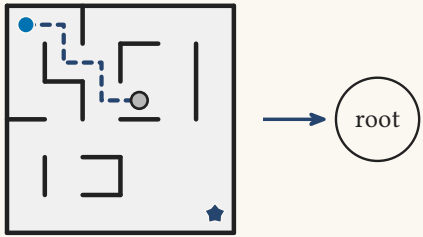
Each state carries N, its passes, and V, their average G.



a state: the door, on turn 0

A counter on every state

Each state carries N, its passes, and V, their average G.

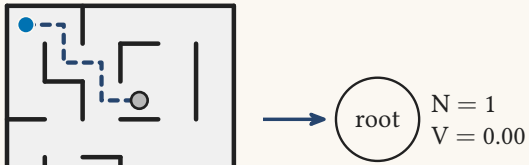


a state: the door, on turn 0

1 A pass is one imagined run through it, here ending in stone: $G = 0$.

A counter on every state

Each state carries N, its passes, and V, their average G.



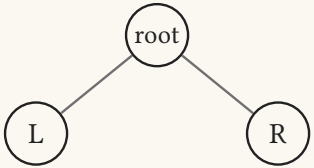
a state: the door, on turn 0

1 A pass is one imagined run through it, here ending in stone: $G = 0$.

2 Born with $N = 1$ and $V = 0$: one imagined visit that paid nothing.

A tree of depth d

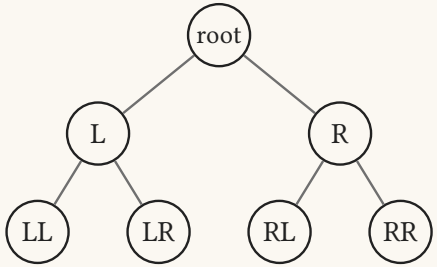
The first version: a node for every state up to d steps ahead.



1 A toy labyrinth: three forks. The door is the root.

A tree of depth d

The first version: a node for every state up to d steps ahead.

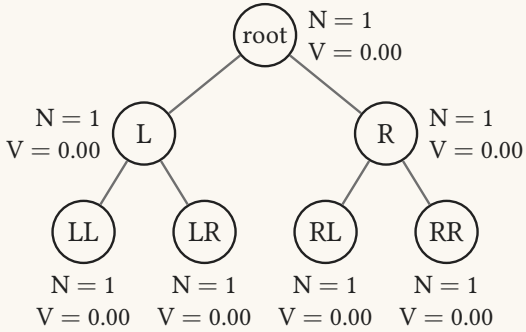


1 A toy labyrinth: three forks. The door is the root.

2 With $d = 2$: 7 nodes.

A tree of depth d

The first version: a node for every state up to d steps ahead.



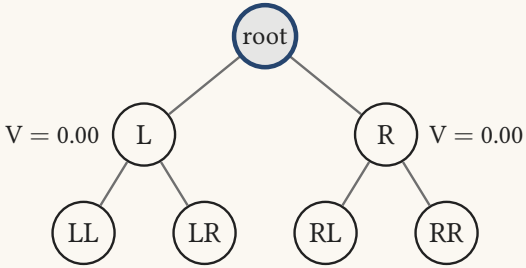
1 A toy labyrinth: three forks. The door is the root.

2 With $d = 2$: 7 nodes.

3 Each starts at $N = 1$, $V = 0$.

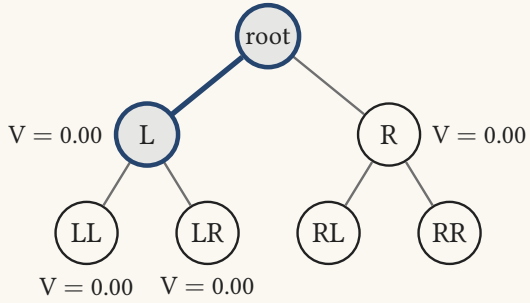
A pass, step 1: go down

A pass goes down, into the child with the largest V.



A pass, step 1: go down

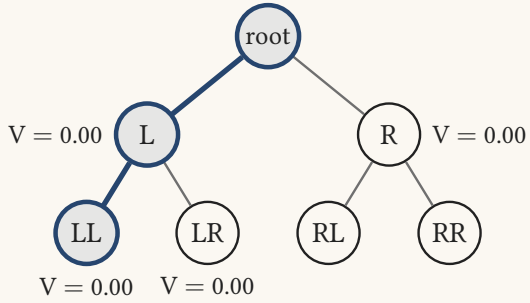
A pass goes down, into the child with the largest V.



1 $V = 0$ twice: a tie, so the left child.

A pass, step 1: go down

A pass goes down, into the child with the largest V.

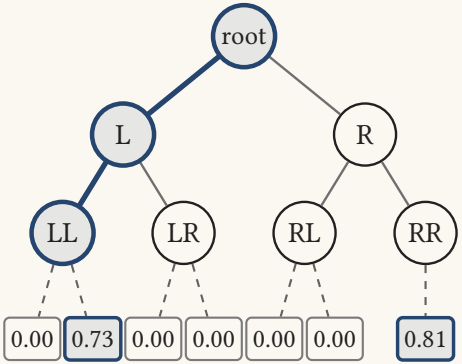


1 $V = 0$ twice: a tie, so the left child.

2 Again, and it stops at LL: the tree ends.

A pass, step 2: a rollout

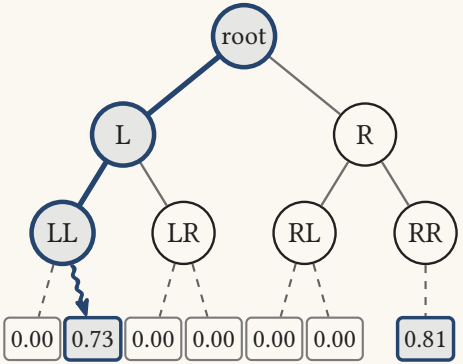
A rollout plays at random to the end and reads off G.



1 In the toy a lair on turn t pays 0.9^t, any other end 0.

A pass, step 2: a rollout

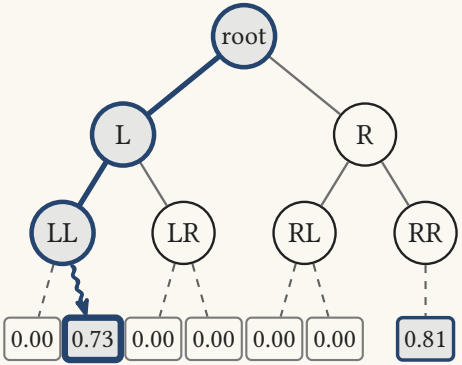
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A pass, step 2: a rollout

A rollout plays at random to the end and reads off G.

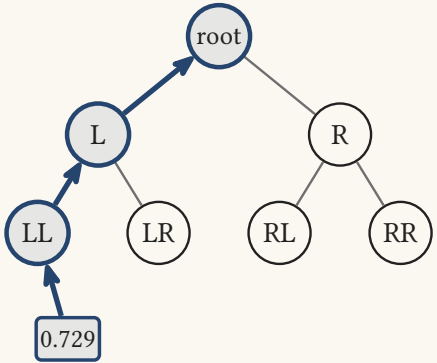


1 In the toy a lair on turn t pays 0.9^t, any other end 0.

2 $G = 0.9^3 = 0.729$. A rollout adds no nodes.

A pass, step 3: walk back up

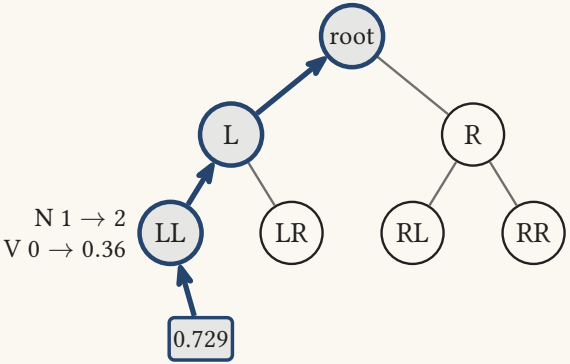
Each node on the path counts the visit and moves V toward G.



1 $G = 0.729$ goes back up the path.

A pass, step 3: walk back up

Each node on the path counts the visit and moves V toward G.

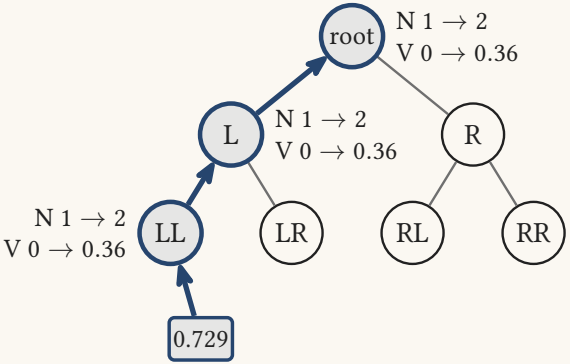


1 $G = 0.729$ goes back up the path.

2 Each node: $N \leftarrow N + 1$, $V \leftarrow V + (G - V)/N$. At LL, $V = 0.00 + (0.729 - 0.00)/2 \approx 0.36$.

A pass, step 3: walk back up

Each node on the path counts the visit and moves V toward G.



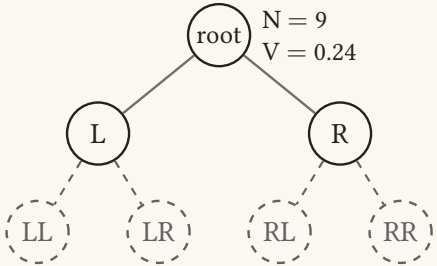
1 $G = 0.729$ goes back up the path.

2 Each node: $N \leftarrow N + 1$, $V \leftarrow V + (G - V)/N$. At LL, $V = 0.00 + (0.729 - 0.00)/2 \approx 0.36$.

3 Same at L and the root; the others are untouched.

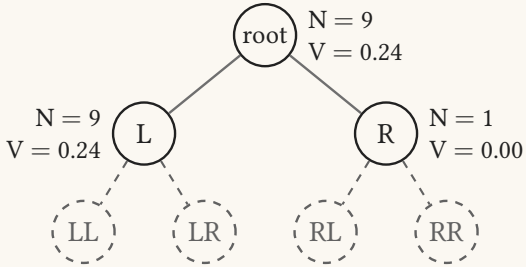
M passes, then one move

After M passes, he moves to the most visited child.



M passes, then one move

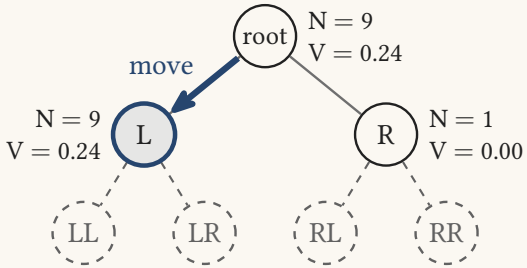
After M passes, he moves to the most visited child.



1 Left has $N = 9$: 8 passes and its birth visit. Right still has $N = 1$.

M passes, then one move

After M passes, he moves to the most visited child.

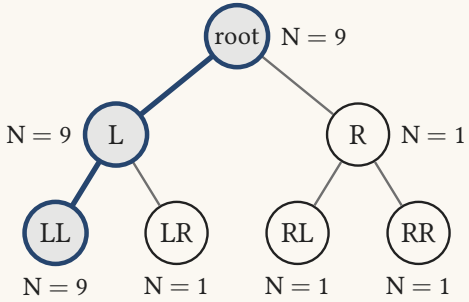


1 Left has $N = 9$: 8 passes and its birth visit. Right still has $N = 1$.

2 He moves into it: left.

When the first success takes over

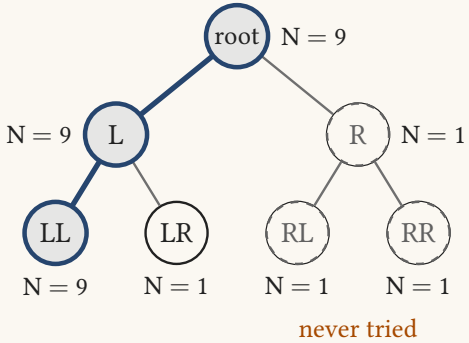
With ties to the first child, the first success keeps every pass.



1 All 8 went left: any V above 0 beats an untried 0.

When the first success takes over

With ties to the first child, the first success keeps every pass.

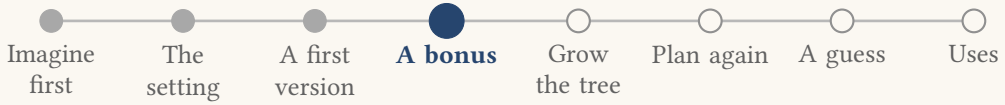


1 All 8 went left: any V above 0 beats an untried 0.

2 Right was never tried, yet it averages 0.41 against 0.18 for left.

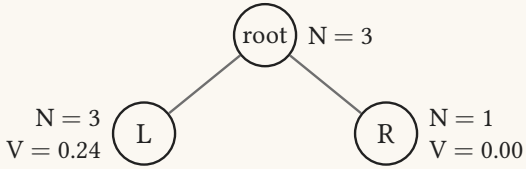
A bonus for the barely tried

Why the first version sticks, and how UCT repairs it.



A child that paid once

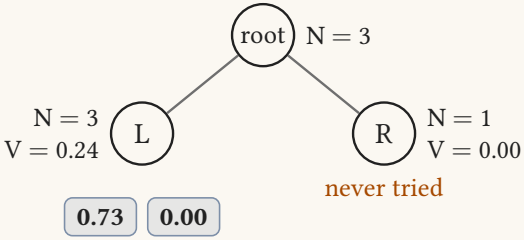
A child that paid once beats one nobody tried, and greedy never goes back.



1 Two passes so far, both through the left child.

A child that paid once

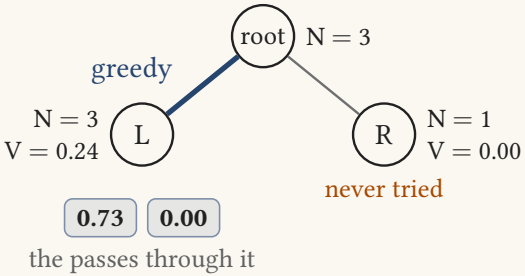
A child that paid once beats one nobody tried, and greedy never goes back.



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A child that paid once

A child that paid once beats one nobody tried, and greedy never goes back.

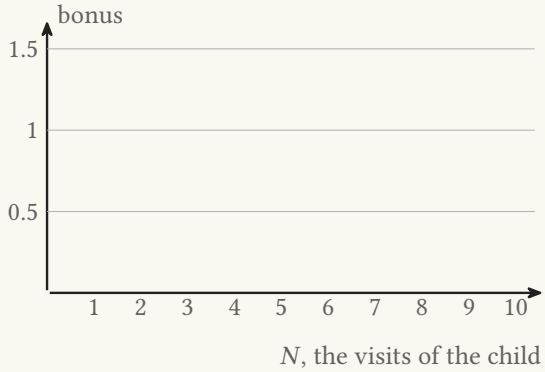


1 Two passes so far, both through the left child.

2 Only a pass through the right could change its V; greedy sends none.

A bonus for the barely tried

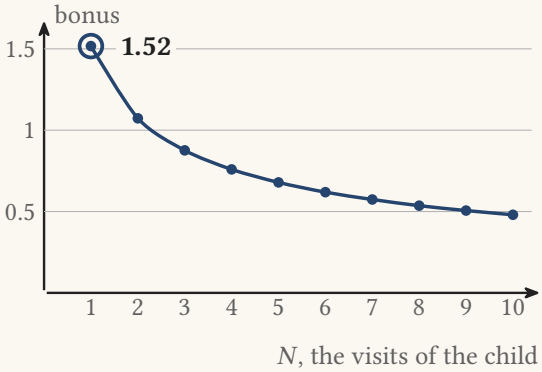
Add to V a bonus that shrinks as the child's visits grow.



1 For a parent visited 10 times.

A bonus for the barely tried

Add to V a bonus that shrinks as the child's visits grow.

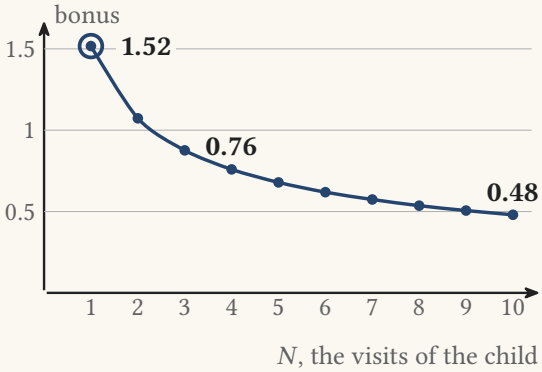


1 For a parent visited 10 times.

2 An untried child, $N = 1$, gets the largest: 1.52.

A bonus for the barely tried

Add to V a bonus that shrinks as the child's visits grow.



1 For a parent visited 10 times.

2 An untried child, $N = 1$, gets the largest: 1.52.

3 It shrinks: 0.76 at $N = 4$, 0.48 at $N = 10$.

UCT

A pass goes into the child with the largest score.

UCT: Upper Confidence bounds applied to Trees.

$$\mathrm{UCT}(s) \;=\; V(s) \;+\; w \cdot \mathrm{bonus}(s)$$

1 w is a weight we choose.

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ln: natural logarithm.

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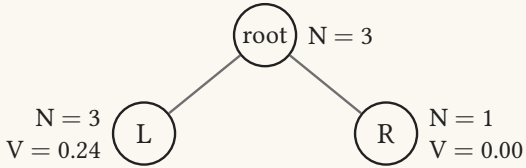
ln: natural logarithm.

1. w is a weight we choose.
2. It is large when $N(s)$ is small next to its parent's.
3. Each pass that skips s raises it, so s gets its turn.

Only the descent changes.

One choice, worked out

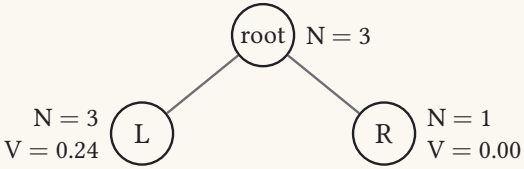
Left has the larger V, the untried right child the larger score.



1 Pass 3, with $w = 1$.

One choice, worked out

Left has the larger V, the untried right child the larger score.

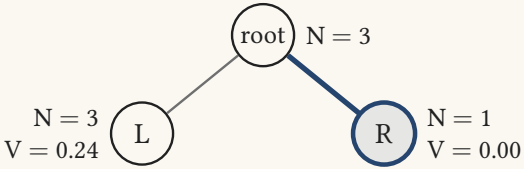


1 Pass 3, with $w = 1$.

	bonus	score $= V +$ bonus
L	$\sqrt{\ln 3 \,/\, 3} = 0.61$	$0.24 + 0.61 = \mathbf{0.85}$
R	$\sqrt{\ln 3 \,/\, 1} = 1.05$	$0.00 + 1.05 = \mathbf{1.05}$

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Left has the larger V, the untried right child the larger score.



	bonus	score $= V +$ bonus
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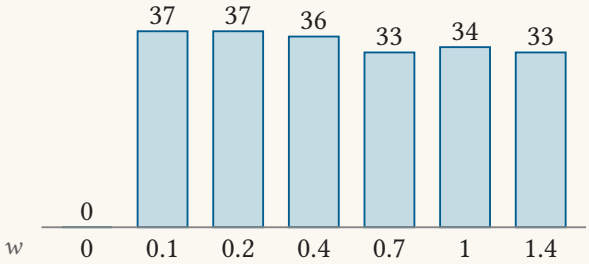
1 Pass 3, with $w = 1$.

2 Right scores 1.05 against 0.85, so pass 3 goes right. Greedy would go left.

The size of w

w, the exploration weight, sets how much the unknown counts.

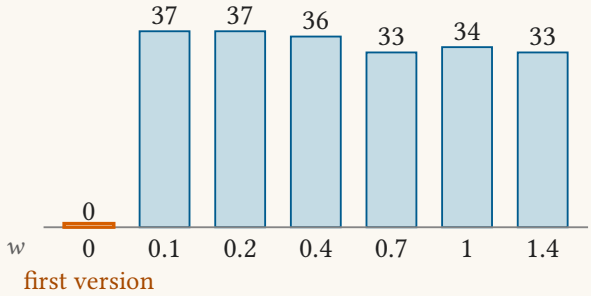
games won of 40; full tree of depth 2; 100 passes a move



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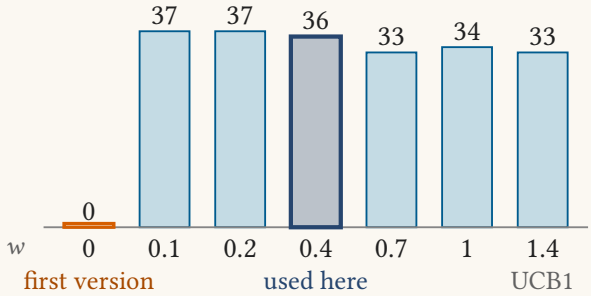


1 The first version, $w = 0$, wins none; any other w wins 33 to 37.

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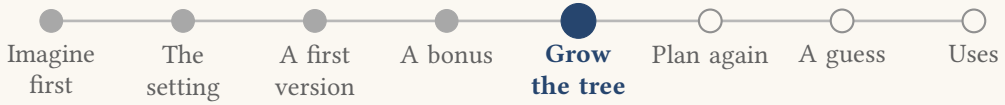
1 The first version, $w = 0$, wins none; any other w wins 33 to 37.

2 UCB1 uses $\sqrt{2} \approx 1.4$; we use 0.4.

Auer, Cesa-Bianchi, Fischer (2002), p. 245: UCB1.

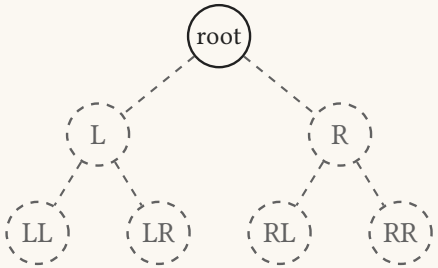
Grow the tree where passes go

One node a pass: select, expand, simulate, back up.



The tree grows only where passes go

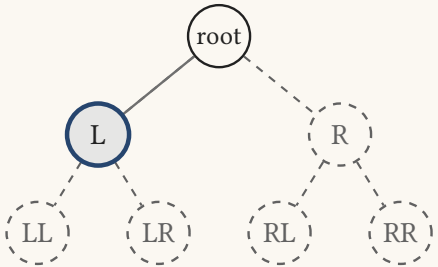
A full tree is mostly never visited: grow one node per pass.



1 At depth 10 its last row has 93 thousand nodes.

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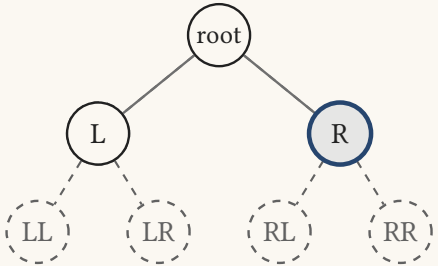


1 At depth 10 its last row has 93 thousand nodes.

2 Pass 1 adds L, pass 2 R, pass 3 LL.

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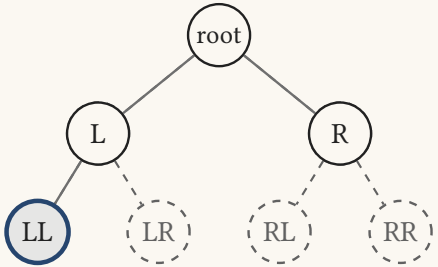


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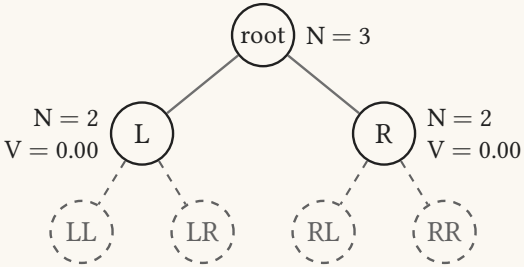
1 At depth 10 its last row has 93 thousand nodes.

2 Pass 1 adds L, pass 2 R, pass 3 LL.

3 Dashed: not built yet.

Where a pass stops

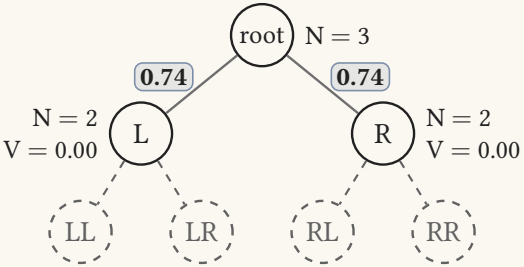
A pass goes down until a node has an untried action.



1 L and R exist; the dashed nodes are untried actions.

Where a pass stops

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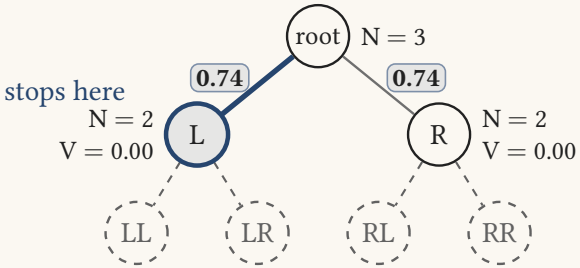


1 L and R exist; the dashed nodes are untried actions.

2 The larger UCT wins: a tie at 0.74, so L.

Where a pass stops

A pass goes down until a node has an untried action.

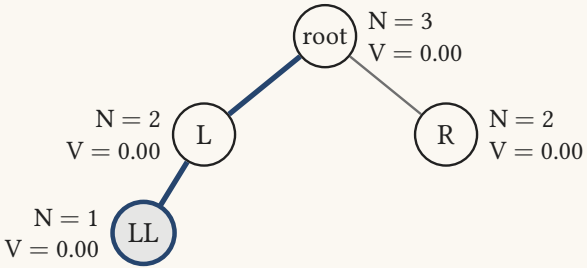


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The new node is the start

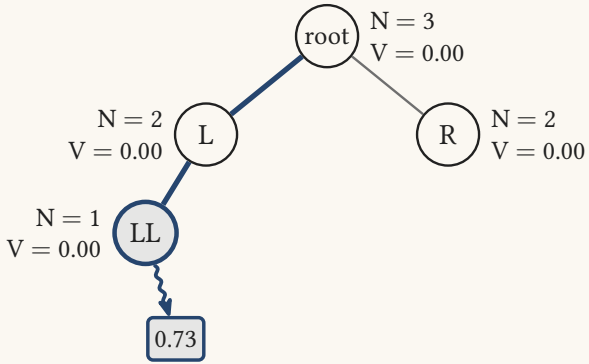
The new child starts the rollout and the walk back up.



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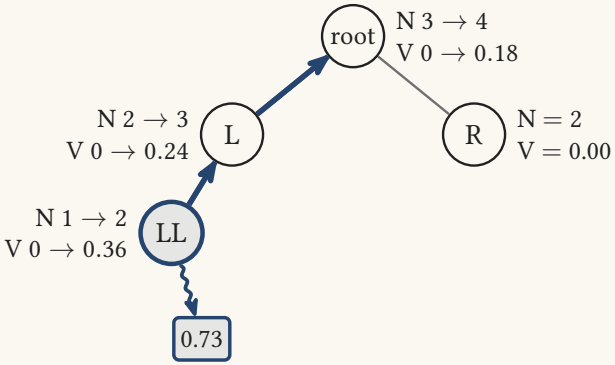


1 It adds LL, born with $N = 1$, $V = 0$.

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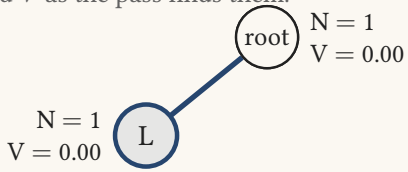
2 The rollout starts at LL and reaches the lair: $G = 0.73$.

3 LL, L and the root count it; R is untouched.

Six passes on the toy

Six passes grow the tree to 7 nodes, UCT at $w = 1$.

N and V as the pass finds them.

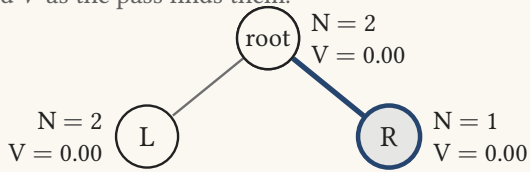


pass	adds	paid	UCT of L / R
1	L	0	*untried*

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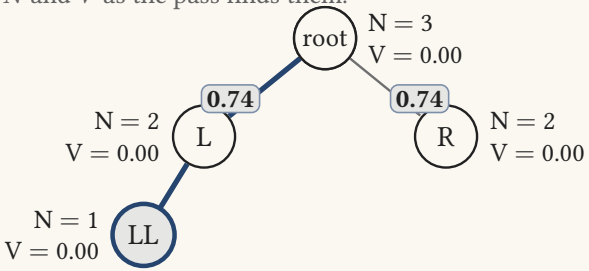


pass	adds	paid	UCT of L / R
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2	R	0	*untried*

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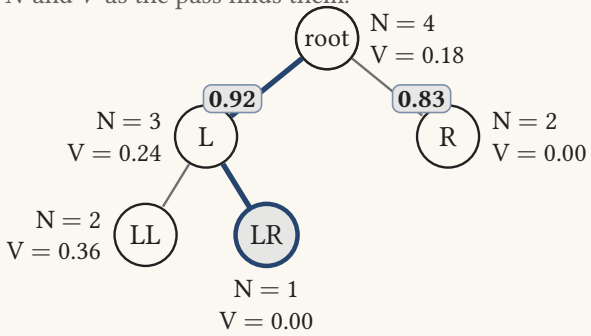
N and V as the pass finds them.



pass	adds	paid	UCT of L / R
1	L	0	*untried*
2	R	0	*untried*
3	LL	0.73	**0.74** / 0.74

Six passes on the toy

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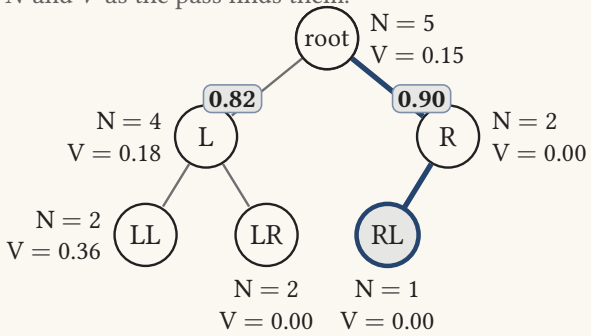


pass	adds	paid	UCT of L / R
1	L	0	*untried*
2	R	0	*untried*
3	LL	0.73	**0.74** / 0.74
4	LR	0	**0.92** / 0.83

Pass 4 stops at L: LR untried.

Six passes on the toy

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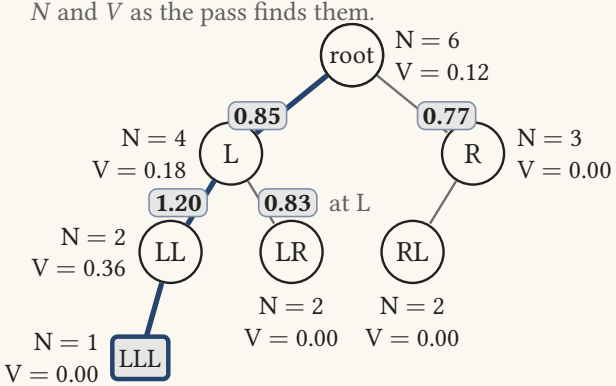


pass	adds	paid	UCT of L / R
1	L	0	*untried*
2	R	0	*untried*
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4	LR	0	**0.92** / 0.83
5	RL	0	0.82 / **0.90**

Pass 4 stops at L: LR untried.
Pass 5: R has the lower V but the higher UCT.

Six passes on the toy

Six passes grow the tree to 7 nodes, UCT at $w = 1$.

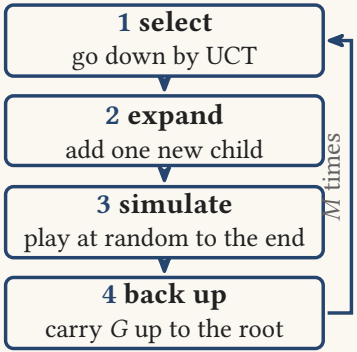


pass	adds	paid	UCT of L / R
1	L	0	*untried*
2	R	0	*untried*
3	LL	0.73	**0.74** / 0.74
4	LR	0	**0.92** / 0.83
5	RL	0	0.82 / **0.90**
6	LLL	0	**0.85** / 0.77

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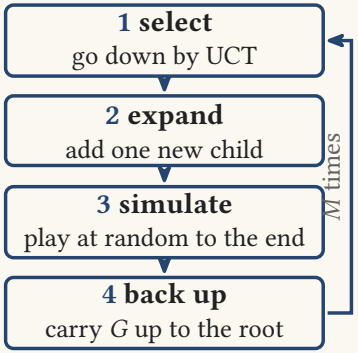
The four phases

Every pass: select, expand, simulate, back up.



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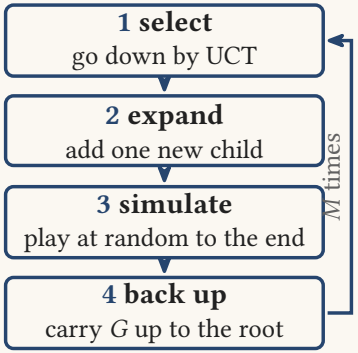


`mcts_move(state, n_attempts, w, rng)`

```
root = Node(state, None)
for i in range(n_attempts):
   leaf = descend(root, w)
   leaf = expand(leaf)
   r = rollout_reward(leaf.state)
   backprop(leaf, r)
return most_visited_child(root)
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4   backprop(leaf, r)
return most_visited_child(root)
```

M is `n_attempts` in the code.

The search in code: select and expand

One function walks down the tree, the other adds a node.

1
```
def descend(s, w):
    if len(s.untried) == 0 and not game_over(s.state):
        best = best_child_by_uct(s, w)
        return descend(best, w)
    else:
        return s
```

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        return descend(best, w)
    else:
        return s
```

2
```
def expand(s):
    if game_over(s.state):
        return s
    action = s.untried.pop(0)
    new_state = env.step(s.state, action)
    child = Node(new_state, s)
    s.children.append(child)
    return child
```

The search in code: select and expand

One function walks down the tree, the other adds a node.

1
```
def descend(s, w):
    if len(s.untried) == 0 and not game_over(s.state):
        best = best_child_by_uct(s, w)                    # largest UCT
        return descend(best, w)
    else:
        return s                                          # stop: untried action, or run over
```

2
```
def expand(s):
    if game_over(s.state):
        return s
    action = s.untried.pop(0)                             # first untried action, removed
    new_state = env.step(s.state, action)
    child = Node(new_state, s)
    s.children.append(child)
    return child                                          # start of rollout and back up
```

The search in code: simulate and back up

One function reads off G, the other carries it up the path.

```
3 def rollout_reward(state, rng):
      final_state = rollout(state, rng)
      return outcome(final_state)
```

The search in code: simulate and back up

One function reads off G, the other carries it up the path.

3
```
def rollout_reward(state, rng):
    final_state = rollout(state, rng)
    return outcome(final_state)
```

4
```
def backprop(node, r):
    node.N = node.N + 1
    node.V = (node.V * (node.N - 1) + r) / node.N
    if node.parent is not None:
        backprop(node.parent, r)
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The search in code: simulate and back up

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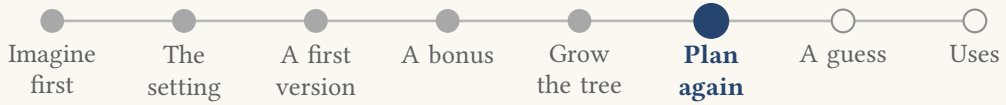
3
```
def rollout_reward(state, rng):
    final_state = rollout(state, rng)          # random play to the end
    return outcome(final_state)                # its return, G
```

4
```
def backprop(node, r):
    node.N = node.N + 1                        # one more visit
    node.V = (node.V * (node.N - 1) + r) / node.N
    if node.parent is not None:
        backprop(node.parent, r)               # on to the parent
```

That is the running mean: $V \leftarrow V + (r - V)/N$.

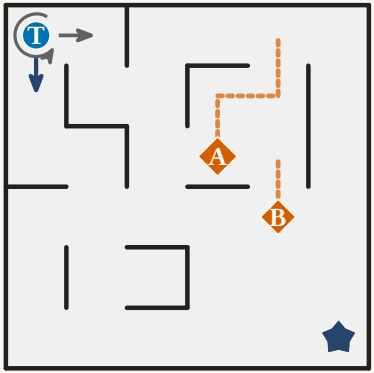
Planning is repeated

One search buys one step; then we search again.

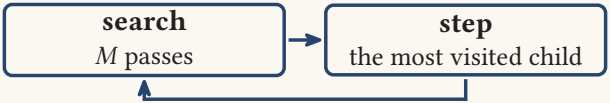


Plan, step, plan again

One search buys one step; then we search again.

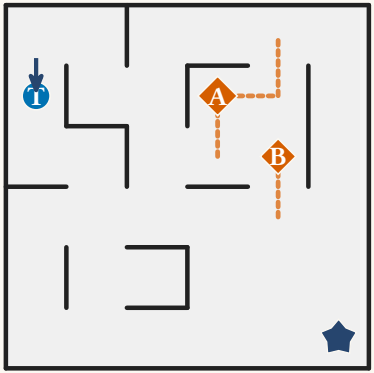


1 At the door, 100 passes: stay $N = 33$, right 32, down 38. He steps down.



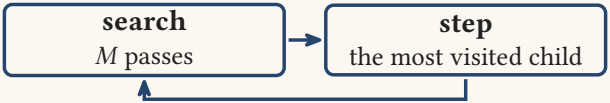
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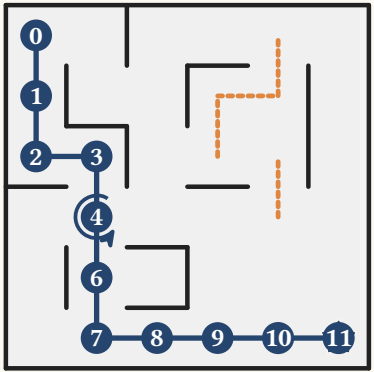
1 At the door, 100 passes: stay $N = 33$, right 32, down 38. He steps down.

2 The Medusas move: search again from here.



Plan, step, plan again

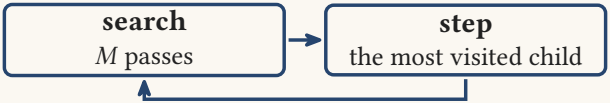
One search buys one step; then we search again.



1 At the door, 100 passes: stay $N = 33$, right 32, down 38. He steps down.

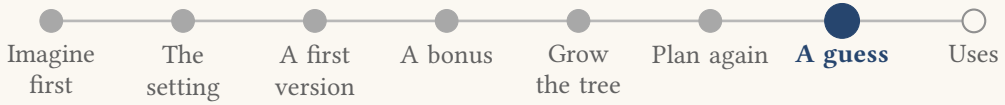
2 The Medusas move: search again from here.

3 11 steps, 11 searches.



Start from a guess

When random rollouts know nothing, a guess helps.



What the rollouts know

Rollouts that pay nothing make the first moves look alike.

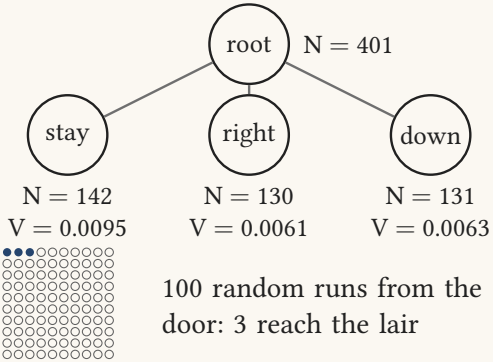
1 From the door, 97.2% of random runs pay nothing.



100 random runs from the door: 3 reach the lair

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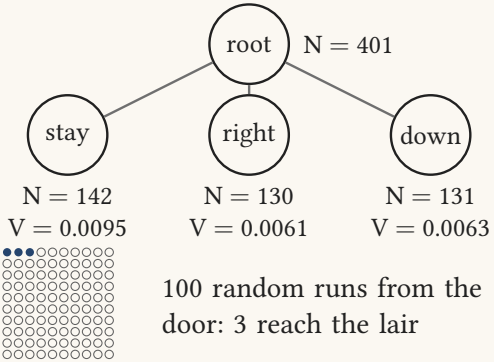
100 random runs from the door: 3 reach the lair

1 From the door, 97.2% of random runs pay nothing.

2 After 400 passes their V differ by only 0.0035.

What the rollouts know

Rollouts that pay nothing make the first moves look alike.



100 random runs from the door: 3 reach the lair

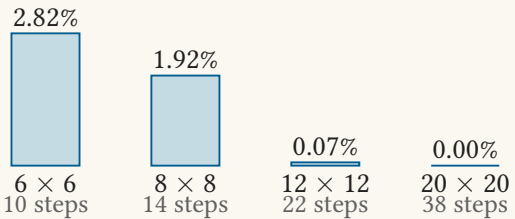
1 From the door, 97.2% of random runs pay nothing.

2 After 400 passes their V differ by only 0.0035.

3 The bonus, about 0.08, is 24 times that, so counts stay level.

The wall

On larger labyrinths random rollouts stop reaching the lair.



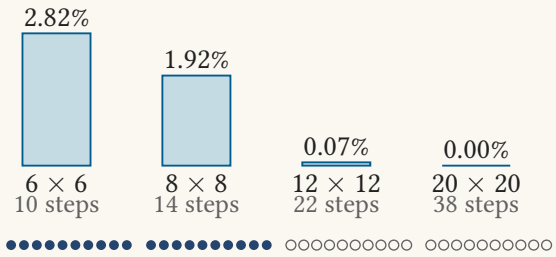
Bars: walks that reach the lair.

1 Bigger labyrinths, same search: $M = 400$.

2 The share falls: 2.82%, 0.07%, then 0.00%.

The wall

On larger labyrinths random rollouts stop reaching the lair.



Bars: walks that reach the lair.
Dots: games won of 10.

1 Bigger labyrinths, same search: $M = 400$.

2 The share falls: 2.82%, 0.07%, then 0.00%.

3 Wins fall too: only zeros to average.

Start from a guess

A new node can start from a guess V_0.

a new node: the move down



imagined

$N = 1, V = 0$

1 Until now: one imagined visit that paid nothing.

Start from a guess

A new node can start from a guess V_0.

a new node: the move down

0.74

imagined
$N = n_0 = 1$, $V = V_0 = 0.74$

the guess: what it pays with no Medusas,
$0.97^{1+9} = 0.74$

1 Until now: one imagined visit that paid nothing.

2 Now: n_0 imagined visits, each paying V_0.

Start from a guess

A new node can start from a guess V_0.

a new node: the move down



imagined real rollouts

$N = 3$, $V = (0.74 + 0 + 0)/3 = 0.25$

the guess: what it pays with no Medusas, $0.97^{1+9} = 0.74$

As before: $N \leftarrow N + 1$, $V \leftarrow V + (G - V)/N$.

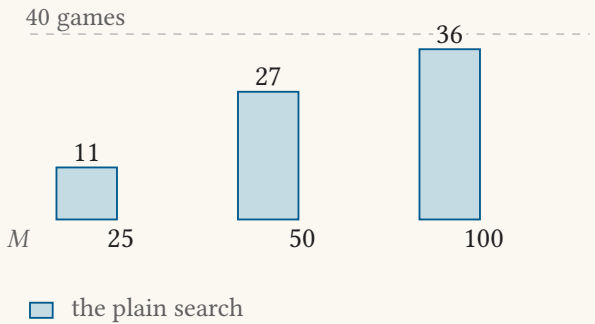
1 Until now: one imagined visit that paid nothing.

2 Now: n_0 imagined visits, each paying V_0.

3 Real rollouts that pay nothing pull it down.

A guess lifts a starved search

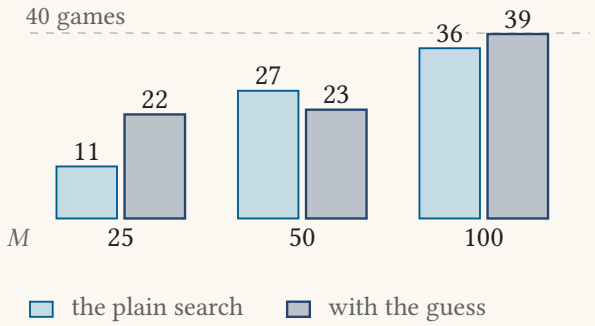
A guess that ignores the Medusas lifts $M = 25$ from 11 to 22 wins of 40.



1 Plain: 11, 27 and 36 wins.

A guess lifts a starved search

A guess that ignores the Medusas lifts $M = 25$ from 11 to 22 wins of 40.

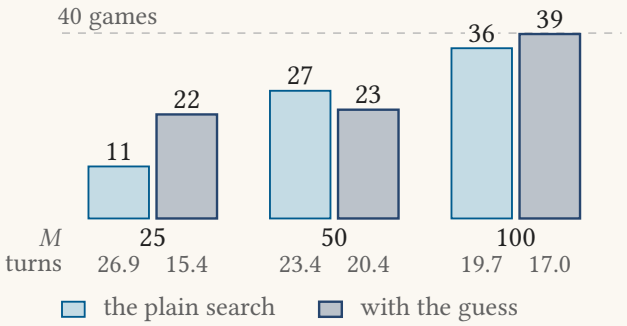


1 Plain: 11, 27 and 36 wins.

2 With the guess: 22, 23, 39: 11 more, 4 fewer, 3 more.

A guess lifts a starved search

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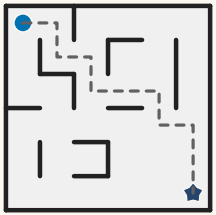
1 Plain: 11, 27 and 36 wins.

2 With the guess: 22, 23, 39: 11 more, 4 fewer, 3 more.

3 Turns of won games fall too.

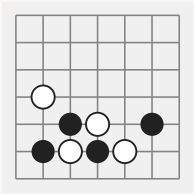
Where guesses come from

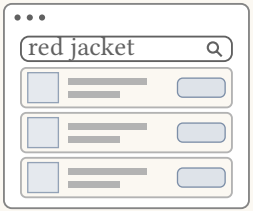
A rule, a network or a language model can supply V_0.



a rule: the walking distance

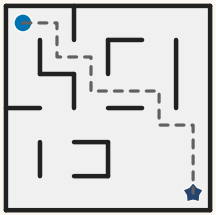
$V_0 = 0.97^{t+d}$





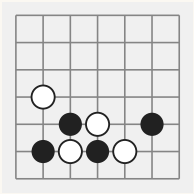
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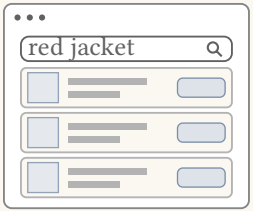
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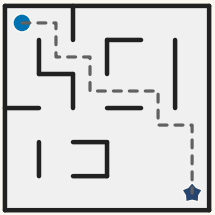
a trained network

its value



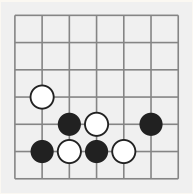
Where guesses come from

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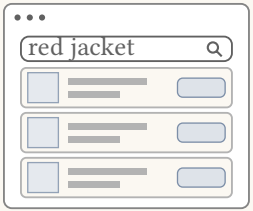
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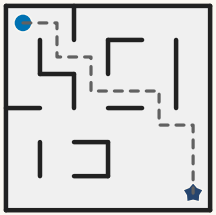


a language model

its score

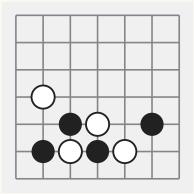
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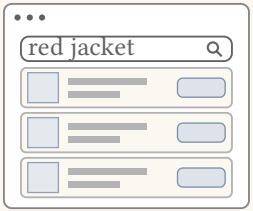
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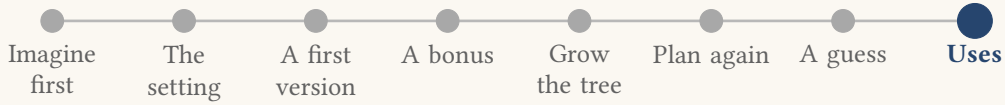
a language model

its score

The guess starts the node; rollouts correct it.

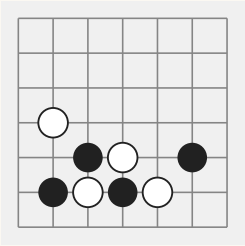
Where it is used

AlphaGo and language-model agents run a version of this loop.



AlphaGo, in the four words

The same four words, for the game of Go.



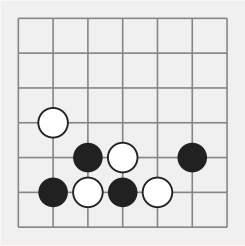
a position

state	the board position
action	a legal move

Silver et al., *Nature* 529, 484 to 489 (2016).

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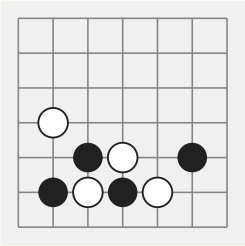
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state	the board position
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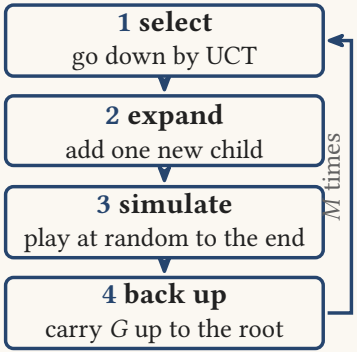
state	the board position
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transition	the rules of Go
return	win or loss

About 250 moves a position, 150 moves a game.

Silver et al., *Nature* 529, 484 to 489 (2016).

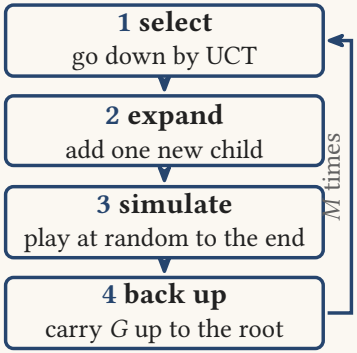
What AlphaGo adds to the loop

AlphaGo keeps the loop and the most visited move; networks feed it.



What AlphaGo adds to the loop

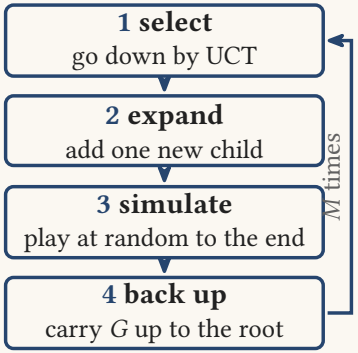
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1. Priors from a policy network scale the bonus.

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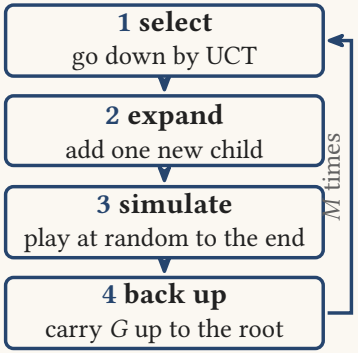
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3. Network value v and fast rollout z score the leaf.

   $V = (1 - \lambda)\, v + \lambda\, z$, with $\lambda = 0.5$

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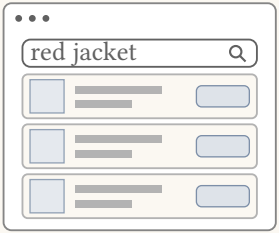
3. Network value v and fast rollout z score the leaf.

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4. Two players: one's win is the other's loss.

LATS, in the four words

The same four words, for a language-model agent.



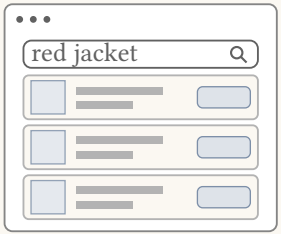
a task on the web

state	the task so far, as text
action	$n = 5$ steps written by the model

Zhou et al., ICML 2024 (PMLR 235).

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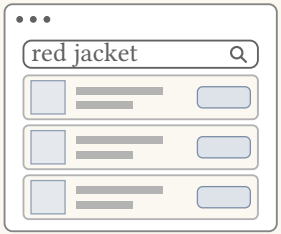
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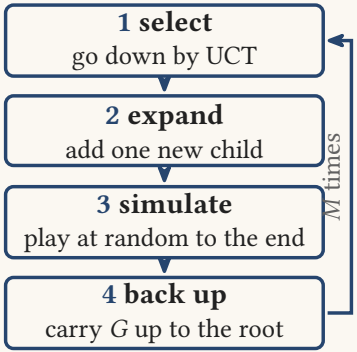
state	the task so far, as text
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transition	the site answers; replay to go back
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Any state is restored by pasting its text back.

Zhou et al., ICML 2024 (PMLR 235).

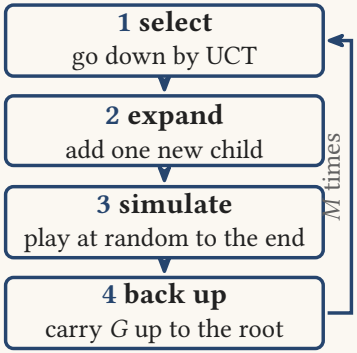
What LATS adds to the loop

LATS puts a language model in the loop, and trains nothing.



What LATS adds to the loop

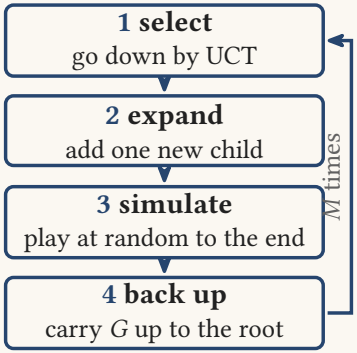
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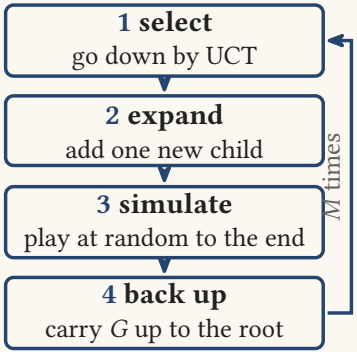
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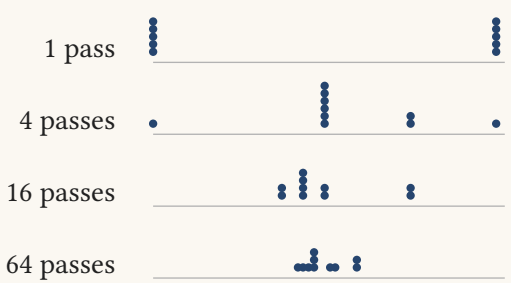
2. The model proposes 5 actions and scores each.
3. The rollout follows the scores.
4. A failed run leaves a written reflection.

Appendix

Extras for questions: where the bonus comes from, and more.

Where the bonus comes from

An average of N passes strays from the truth like $1/\sqrt{N}$.

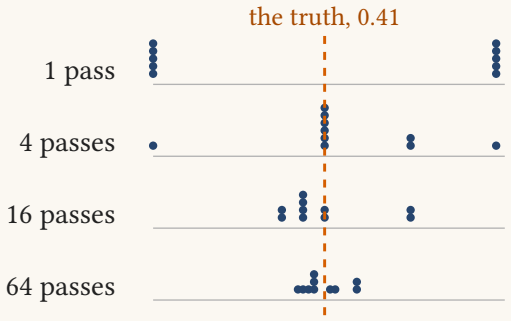


1 Right fork: a pass pays 0.81 or 0, equally likely. Each row: 10 averages of N passes.

Auer, Cesa-Bianchi, Fischer (2002), pp. 237, 245.

Where the bonus comes from

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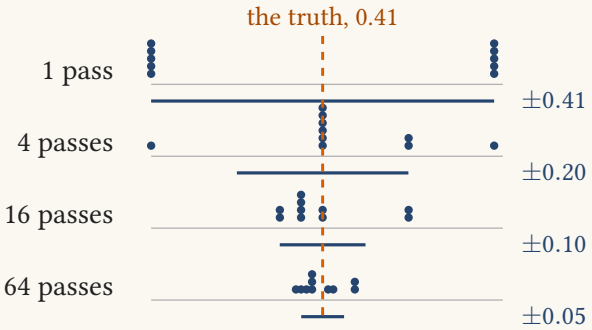
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2 More passes crowd round the truth.

Where the bonus comes from

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1 Right fork: a pass pays 0.81 or 0, equally likely. Each row: 10 averages of N passes.

2 More passes crowd round the truth.

3 The bar: how far an average typically strays, half as far when N quadruples.

From a confidence bound to the bonus

A confidence bound on an average, rewritten, is the bonus of UCT.

1. n plays of a child, each G in $[0, 1]$: average $\bar{G}$, true mean μ.

Hoeffding (1963); Kocsis and Szepesvári (2006), eqs. 2, 3.

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3. Set the right side to δ: $\varepsilon = \sqrt{\ln(1/\delta)/(2n)}$; for $\delta = 0.05$, $n = 10$ it is 0.39.
4. Take $\delta = N^{-4}$, N the parent's passes: $\varepsilon = \sqrt{2 \ln N / n}$.

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1. n plays of a child, each G in $[0, 1]$: average $\bar{G}$, true mean μ.
2. Hoeffding: $\Pr(\mu \geq \bar{G} + \varepsilon) \leq e^{-2n\varepsilon^2}$.
3. Set the right side to δ: $\varepsilon = \sqrt{\ln(1/\delta)/(2n)}$; for $\delta = 0.05$, $n = 10$ it is 0.39.
4. Take $\delta = N^{-4}$, N the parent's passes: $\varepsilon = \sqrt{2 \ln N / n}$.
5. Write the constant $\sqrt{2}$ as a weight w we tune: the bonus is $w\sqrt{\ln N / n}$.

Hoeffding (1963); Kocsis and Szepesvári (2006), eqs. 2, 3.

The update, step by step

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Node L of the toy: pass 1 pays 0.729, so $N = 2$ and $V = 0 + (0.729 - 0)/2 = 0.36$. Pass 2 pays 0, so $N = 3$ and $V = 0.36 + (0 - 0.36)/3 = 0.24$.

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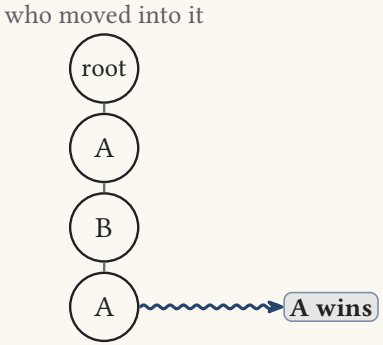
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Starting at $N = 0$ divides the bonus by 0, so each child needs a first try.

Two players, one tree

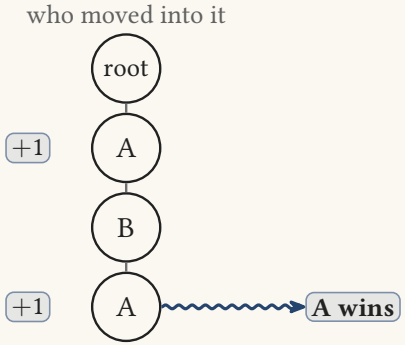
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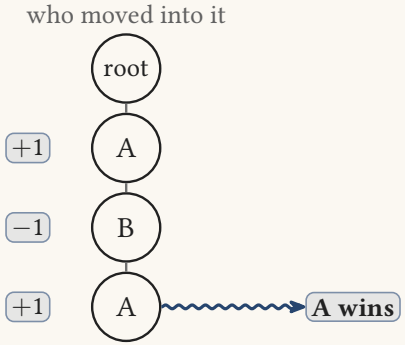


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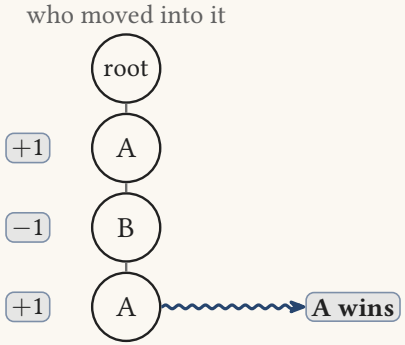
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3 So the sign flips at every step up.

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What UCT is proved to do

The proof gives convergence, and says little about speed or the weight.

Proved

The chance of a wrong move at the root tends to 0, at a polynomial rate.

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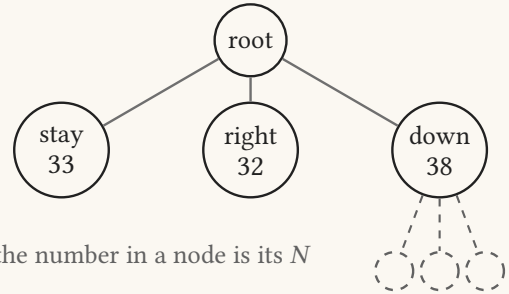
The best w: here we tuned it, 0.4.

Speed: in some trees UCT is over-optimistic, with very poor worst-case regret.

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Reusing the subtree

After the step, planners keep what the search already learned.

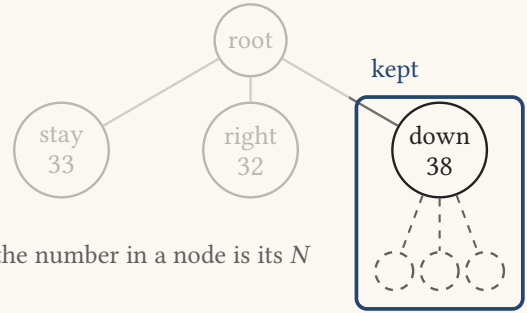


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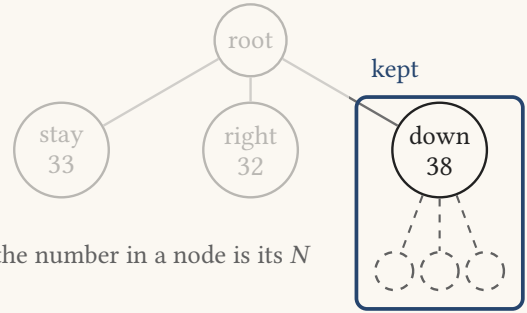
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3 Here each step starts from a new tree: simpler.